

# Chapter 3 Paper 2 — QP + MS Direct Screenshots

Updated to ignore/replace the oldest selected years: this pack uses 2017–2025 Paper 2 questions only. Both QP pages and matching MS pages are direct screenshots from repository PDFs.

| Syllabus area                                                                       | Covered in sections |
|-------------------------------------------------------------------------------------|---------------------|
| 3.1 wave motion/features/equation                                                   | 1, 2, 4, 5, 8, 9    |
| Transverse/longitudinal waves                                                       | 1, 2, 3, 5, 8, 9    |
| Ripple tank reflection/refraction/diffraction                                       | 1, 4, 5, 8, 9       |
| 3.2.1 reflection of light / plane mirror                                            | 7, 9                |
| 3.2.2 refraction, refractive index, critical angle, TIR, optical fibres-related TIR | 3, 6, 8             |

**1) 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank refraction, diffraction — QP screenshot**

- 5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.

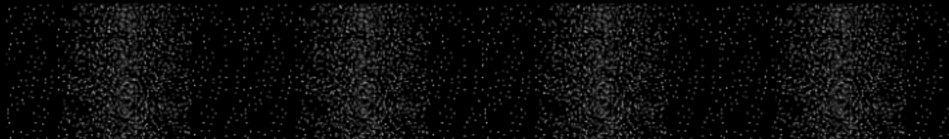


Fig. 5.1

- (i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]
- (ii) Describe the difference between a compression and a rarefaction.
- .....
- .....
- ..... [1]
- (b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.
- (i) The wooden bar makes 45 complete oscillations in 1.0 minute.
- Calculate the frequency of the wave produced.

frequency = .....Hz [1]

- (ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed .....

wavelength .....

[2]



**1) 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank refraction, diffraction — QP screenshot**



(iii) The crests of the water wave move into the shallow region shown in Fig. 5.2.

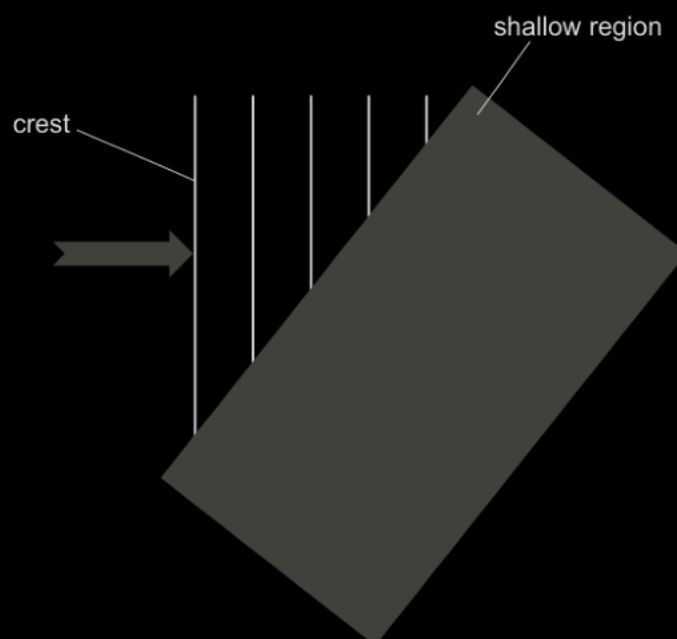


Fig. 5.2

On Fig. 5.2, draw the crests in the shallow region.

[2]

(c) Describe what is meant by the diffraction of a water wave.

.....

.....

..... [1]

[Total: 8]



# 1) 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank refraction, diffraction — matching MS screenshot

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| Question  | Answer                                                                                                                 | Marks |
|-----------|------------------------------------------------------------------------------------------------------------------------|-------|
| 5(a)(i)   | C and R marked correctly                                                                                               | B1    |
| 5(a)(ii)  | compressions are where pressure is high / particles close together / density is high                                   | B1    |
| 5(b)(i)   | 0.75 (Hz)                                                                                                              | B1    |
| 5(b)(ii)  | <i>speed</i> : no change                                                                                               | B1    |
|           | <i>wavelength</i> : decreases                                                                                          | B1    |
| 5(b)(iii) | at least 3 wavefronts in the shallow region parallel to each other <b>and</b> on correct side of normal                | M1    |
|           | refracted towards the surface on correct side of surface <b>and</b> join wavefronts shown                              | A1    |
| 5(c)      | spreading out / bending at an edge / obstacle<br><b>or</b><br>spreading out after passing through a (small) gap / hole | B1    |

2) 5054/21/O/N/24 Q5 — ultrasound, longitudinal vs transverse, wavelength — QP  
screenshot



- 5 Ultrasound is a longitudinal wave which cannot be heard by humans as the frequency of ultrasound is too high.

- (a) Ultrasound waves **cannot** travel in a vacuum.

Explain why.

.....  
.....  
..... [2]

- (b) Describe how a longitudinal wave differs from a transverse wave.

.....  
.....  
..... [2]

- (c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m/s.

- (i) Calculate the wavelength of this ultrasound wave in soft human tissue.

wavelength = ..... m [3]

- (ii) The ultrasound wave passes from the soft tissue into bone where the speed of the ultrasound wave is greater than 1500 m/s.

State what happens to the frequency and to the wavelength of the ultrasound wave as it passes into the bone.

frequency .....

wavelength .....

[2]

[Total: 9]





2) 5054/21/O/N/24 Q5 — ultrasound, longitudinal vs transverse, wavelength — matching MS screenshot

| Question | Answer                                                       | Marks |
|----------|--------------------------------------------------------------|-------|
| 4(c)(ii) | frequency      increases<br>and<br>wavelength      decreases | B1    |
| 4(d)     | infrared (radiation)                                         | B1    |

| Question | Answer                                                                                       | Marks |
|----------|----------------------------------------------------------------------------------------------|-------|
| 5(a)     | ultrasound / sound / it is transmitted by <u>vibrating</u> matter / particles                | B1    |
|          | no matter / particles in a vacuum / no medium                                                | B1    |
| 5(b)     | (in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction  | B1    |
|          | in a transverse wave, vibration (of particles) is perpendicular to the propagation direction | B1    |
| 5(c)(i)  | $(\lambda =) v / f$ or $1500 / 8.4 \times 10^6$                                              | C1    |
|          | $1.8 \times 10^M$                                                                            | C1    |
|          | $1.8 \times 10^{-4}$ (m)                                                                     | A1    |
| 5(c)(ii) | frequency:      does not change                                                              | B1    |
|          | wavelength:      increases                                                                   | B1    |

**3) 5054/21/O/N/23 Q4-Q5 — light refraction + transverse/longitudinal comparison — QP screenshot**

- 4 Red light of frequency  $4.7 \times 10^{14}$  Hz travels in air at a speed of  $3.0 \times 10^8$  m/s.

(a) Calculate the wavelength in air of this red light.

wavelength = ..... m [3]

- (b) Fig. 4.1 shows a narrow beam of this red light striking the surface of a parallel-sided glass block.

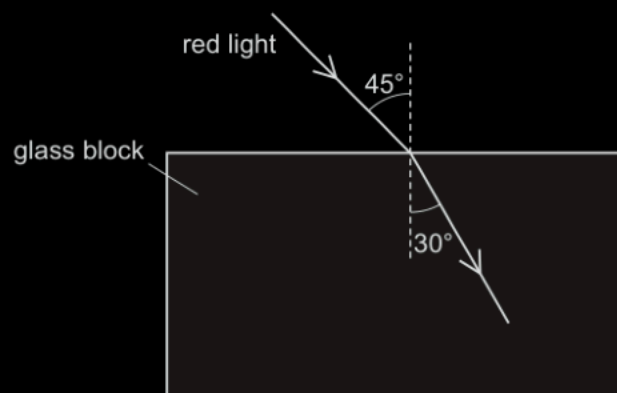


Fig. 4.1

- (i) State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block.

speed .....

frequency .....

wavelength .....

[2]

- (ii) Using the angles shown on Fig. 4.1, calculate the refractive index of the glass.

refractive index = ..... [3]

**3) 5054/21/O/N/23 Q4-Q5 — light refraction + transverse/longitudinal comparison — QP screenshot**

- (iii) The light continues along the path shown in Fig. 4.1 until it strikes the bottom surface of the block. Light then emerges into the air.

Draw on Fig. 4.1 to show the path taken by the light until it strikes the bottom surface and the path of the light that emerges into the air. [2]

[Total: 10]

3) 5054/21/O/N/23 Q4-Q5 — light refraction + transverse/longitudinal comparison — QP screenshot

5 Most of the energy emitted by the Sun is in three regions of the electromagnetic spectrum.

(a) The name of one of these three regions is the ultraviolet region.

(i) State the names of all three regions in order of increasing frequency.

smallest frequency .....

intermediate frequency .....

largest frequency .....

[2]

(ii) Some of the ultraviolet radiation emitted by the Sun reaches the surface of the Earth.

Give one possible damaging effect of ultraviolet radiation on the human body and the property of ultraviolet radiation that causes this damage.

damaging effect .....

.....

property .....

.....

[2]

(b) Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

.....

.....

..... [2]

[Total: 6]

### 3) 5054/21/O/N/23 Q4-Q5 — light refraction + transverse/longitudinal comparison — matching MS screenshot

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| Question  | Answer                                                                                      | Marks |
|-----------|---------------------------------------------------------------------------------------------|-------|
| 4(a)      | $(\lambda =) v / f$ <b>or</b> $3.0 \times 10^8 / 4.7 \times 10^{14}$                        | C1    |
|           | $6.4 \times 10^{-4}$ (m) <b>or</b> $3 / 4.7 \times 10^6$                                    | C1    |
|           | $6.4 \times 10^{-7}$ (m)                                                                    | A1    |
| 4(b)(i)   | speed: <b>decreases</b><br>frequency: <b>stays the same</b><br>wavelength: <b>decreases</b> | B2    |
| 4(b)(ii)  | $i = 45^\circ$ <b>and</b> $r = 30^\circ$                                                    | C1    |
|           | $(n =) \sin(i) / \sin(r)$ <b>or</b> $\sin(45^\circ) / \sin(30^\circ)$                       | C1    |
|           | 1.4                                                                                         | A1    |
| 4(b)(iii) | straight continuation of ray in block <b>and</b> refraction away from normal in air         | B1    |
|           | emergent ray parallel to initial incident ray                                               | B1    |

| Question | Answer                                                                                                      | Marks |
|----------|-------------------------------------------------------------------------------------------------------------|-------|
| 5(a)(i)  | infrared (radiation) <b>and</b> visible (light) <b>and</b> ultraviolet (radiation)                          | B1    |
|          | infrared (radiation) <b>and</b> visible (light) <b>and</b> ultraviolet (radiation) <b>and</b> in this order | B1    |
| 5(a)(ii) | damaging effect: (skin) cancer / cataracts                                                                  | B1    |
|          | property: it / ultraviolet radiation is ionising                                                            | B1    |



### 3) 5054/21/O/N/23 Q4-Q5 — light refraction + transverse/longitudinal comparison — matching MS screenshot

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| Question | Answer                                                                                                               | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------|-------|
| 5(b)     | in a transverse wave, the <u>vibration</u> direction is perpendicular / at right angles to the propagation direction | B1    |
|          | in a longitudinal wave, the <u>vibration</u> direction is parallel to the propagation direction                      | B1    |

| Question | Answer                                                                  |    |                                      | Marks |
|----------|-------------------------------------------------------------------------|----|--------------------------------------|-------|
| 6(a)     | (because) the current at every point in a series circuit is the same    |    |                                      | B1    |
| 6(b)(i)  | $(R =) V / I$ or 1.5 / 0.20 or 7.5 ( $\Omega$ )                         |    |                                      | C1    |
|          | $(R_T =) 7.5 - 1.5 = 3.5$                                               |    |                                      | C1    |
|          | 2.5 ( $\Omega$ )                                                        |    |                                      | A1    |
| 6(b)(ii) | resistance (of T / thermistor) decreases (as the temperature increases) |    |                                      | B1    |
|          |                                                                         |    |                                      |       |
|          | current in circuit increases                                            | or | smaller fraction of total resistance | B1    |
|          |                                                                         |    |                                      |       |
|          | p.d. across R and S increases                                           | or | (p.d. is) smaller fraction of e.m.f. | B1    |

| Question | Answer                                                                                                                 | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------|-------|
| 7(a)(i)  | P: (carbon) brush <b>and</b> Q: slip ring(s)                                                                           | B1    |
| 7(a)(ii) | to connect the output terminal(s) and the (rotating) coil/brushes                                                      | B1    |
|          | without wires twisting                                                                                                 | B1    |
| 7(b)(i)  | at least one of the five positions when the output e.m.f. is zero indicated with X <b>and</b> no indications elsewhere | B1    |

4) 5054/21/M/J/23 Q7 — diffraction through a gap + wavelength definition — QP  
screenshot

- 7 A water wave in a ripple tank diffracts as it passes through the gap in a barrier.

Fig. 7.1 shows a drawing made by a student of the crests in the pattern.

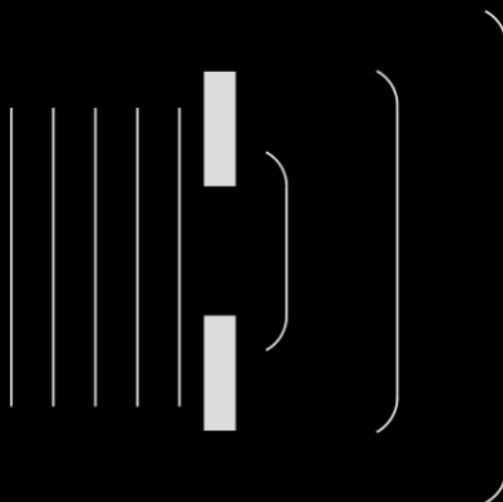


Fig. 7.1

- (a) State **one** way in which the student's drawing of the crests is wrong.

.....  
 ..... [1]

- (b) The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.

Complete Fig. 7.2 with at least three crests to show the new diffraction pattern.



Fig. 7.2

[2]

#### 4) 5054/21/M/J/23 Q7 — diffraction through a gap + wavelength definition — matching MS screenshot

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| Question  | Answer                                                                                                                                                         | Marks |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 7(a)      | wavelength has altered / too large<br>or crests in shadow should be bent / quarter circular above or below gap<br>or vertical lines too straight on two crests | B1    |
| 7(b)      | at least 3 semicircles centered on the gap                                                                                                                     | B1    |
|           | wavelength constant and the same on right and left of gap <b>and</b> some bending                                                                              | B1    |
| 7(c)(i)   | distance between two (consecutive) identical points such as two (consecutive) crests                                                                           | B1    |
| 7(c)(ii)  | $v = f\lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz)                                                                                                    | C1    |
|           | 3(.0) Hz                                                                                                                                                       | A1    |
| 7(c)(iii) | 1.3 (cm)                                                                                                                                                       | B1    |

5) 5054/22/M/J/23 Q5 — transverse examples and diffraction at an edge — QP  
screenshot

5 Water waves are transverse waves.

(a) Underline **two** other examples of transverse waves.

seismic P-waves

seismic S-waves

sound

X-rays

[1]

(b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

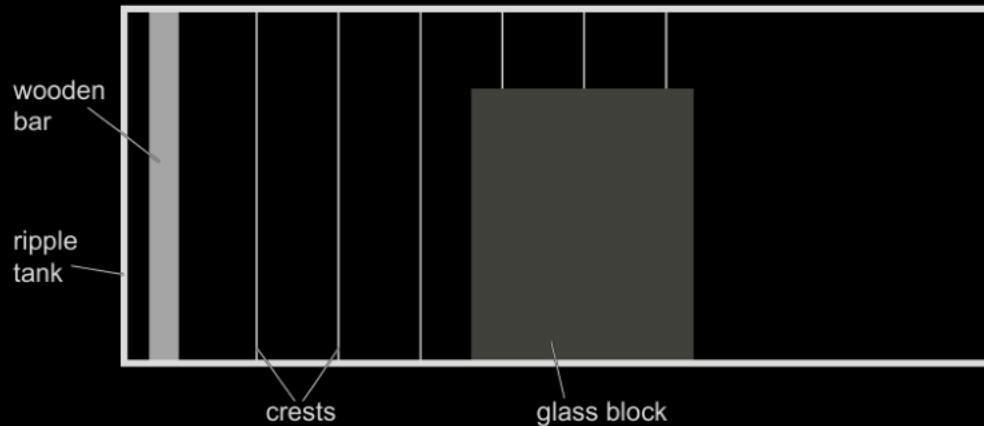


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

(i) The speed of the water wave is 27 cm/s.

Calculate the frequency and the wavelength of the wave.

frequency = ..... Hz

wavelength = ..... cm

[3]

(ii) The wave diffracts at the right-hand edge of the glass block.

On Fig. 5.1 draw **two** crests after they pass the glass block to show the diffraction. [2]

(iii) Describe how a wave with a smaller wavelength is made with the wooden bar.

.....

..... [1]

(iv) Describe how a decrease in wavelength affects the diffraction.

.....

..... [1]

## 5) 5054/22/M/J/23 Q5 — transverse examples and diffraction at an edge — matching MS screenshot

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| Question  | Answer                                                  | Marks |
|-----------|---------------------------------------------------------|-------|
| 4(a)(i)   | –273 (°C) <b>and</b> 0 (K)                              | B1    |
| 4(a)(ii)  | particles) move fast(er)                                | B1    |
|           | <u>vibrate</u> / <u>oscillate</u> with larger amplitude | B1    |
| 4(a)(iii) | internal energy increases                               | B1    |
|           | temperature stays the same                              | B1    |
| 4(b)(i)   | ( $E = mcT$ ) in any form numerical or algebraic        | C1    |
|           | 8000 (J)                                                | A1    |
| 4(b)(ii)  | 100–44 <b>or</b> 56                                     | C1    |
|           | 0.48 <b>or</b> 0.47 (J/(g °C))                          | A1    |

| Question  | Answer                                                                                        | Marks |
|-----------|-----------------------------------------------------------------------------------------------|-------|
| 5(a)      | seismic S-waves <b>and</b> X-rays                                                             | B1    |
| 5(b)(i)   | (wavelength = ) $v/f$ in any form <b>or</b> (distance =) $vt$ <b>or</b> $f = 1/T$ or $1/0.15$ | C1    |
|           | frequency 6.7 (Hz)                                                                            | A1    |
|           | wavelength 4.0 (cm)                                                                           | A1    |
| 5(b)(ii)  | <u>straight</u> wavefronts above block to the right with same wavelength                      | B1    |
|           | correct diffraction – wavefronts quarter circles with centre at top corner                    | B1    |
| 5(b)(iii) | increase frequency of movement of (wooden) bar                                                | B1    |

## 5) 5054/22/M/J/23 Q5 — transverse examples and diffraction at an edge — matching MS screenshot

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| Question | Answer                                  | Marks |
|----------|-----------------------------------------|-------|
| 5(b)(iv) | less diffraction <b>or</b> less bending | B1    |

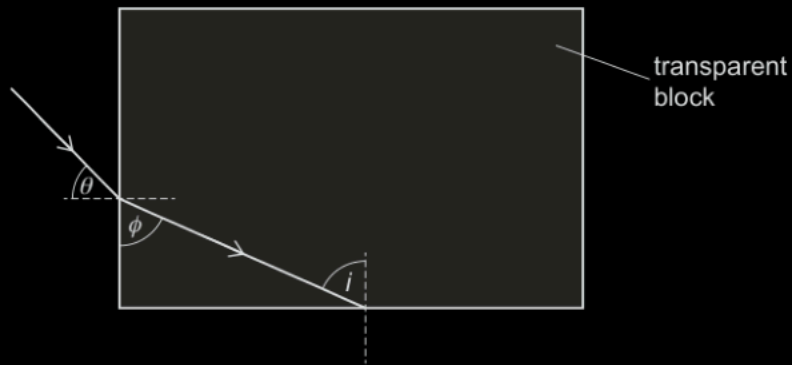
| Question  | Answer                                                                       | Marks |
|-----------|------------------------------------------------------------------------------|-------|
| 6(a)(i)   | ammeter $A_2$ and $A_3$ both 0.25 (A)                                        | B1    |
|           | voltmeter $V_2$ 9(.0 V)                                                      | B1    |
| 6(a)(ii)  | ( $R =$ ) $V / I$ in any form                                                | C1    |
|           | 36 ( $\Omega$ )                                                              | A1    |
| 6(a)(iii) | current is (directly) proportional to voltage / potential difference         | B1    |
|           | provided the temperature is constant                                         | B1    |
| 6(b)      | 0.35 (A) seen                                                                | B1    |
|           | <b>or</b> total resistance $36 + 6 / 0.35$ ; $36 + 17(.1)$ ; 53(.1) $\Omega$ |       |
|           | voltage (across R) $0.35 \times 36$ <b>or</b> 12.6 (V) seen                  | B1    |
|           | 19 V                                                                         | B1    |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                          | Marks |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 7(a)(i)   | X-rays <b>and</b> gamma (radiation / rays)                                                                                                                                                                                                                                                                                                      | B1    |
| 7(a)(ii)  | security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation | B1    |
| 7(a)(iii) | (skin) cancer / cataracts / burns or destroys the retina or cornea                                                                                                                                                                                                                                                                              | B1    |



6) 5054/22/O/N/21 Q3 — refraction, refractive index, critical angle/TIR — QP  
screenshot

- 3 Fig. 3.1 shows light entering a transparent block.



**Fig. 3.1** (not to scale)

The light enters the block at an angle  $\theta$  to the normal and travels through the block until it meets the bottom surface.

The angle between the ray in the block and the vertical side of the block is  $\phi$ .

- (a) Light travels more slowly in the block than in air.

- (i) Explain how Fig. 3.1 shows this.

.....  
 .....  
 ..... [2]

- (ii) State what happens to the wavelength of the light and what happens to the frequency of the light as it enters the block.

wavelength .....  
 frequency .....  
 [2]

- (b) The refractive index of the transparent material is 1.6. Angle  $\theta$  is  $45^\circ$ .

- (i) Determine angle  $\phi$ .

$\phi =$  ..... [3]

6) 5054/22/O/N/21 Q3 — refraction, refractive index, critical angle/TIR — QP  
screenshot

- (ii) The angle of incidence  $i$  at the bottom surface is equal to  $\phi$  and the critical angle for the material of the block in air is  $39^\circ$ .

Explain what happens to the light after it meets the bottom surface.

.....

.....

..... [2]

[Total: 9]

## 6) 5054/22/O/N/21 Q3 — refraction, refractive index, critical angle/TIR — matching MS screenshot

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| Question | Answer                                                                                                              | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------|-------|
| 2(b)(ii) | (water) resistance / resistive force / friction (on swimmer)                                                        | B1    |
|          | (water) resistance / backward force / resistive force / friction / opposing force increases (with speed / time)     | B1    |
|          | (at constant speed) forward force is equal to resistive / backward force<br>or no resultant force or forces balance | B1    |

| Question | Answer                                                                                                                                                                                    | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 3(a)(i)  | it / light refracts / bends / moves towards normal or angle of incidence greater than angle of refraction                                                                                 | B1    |
|          | change in direction of the ray / refraction shows a change in speed (at the boundary) or $\sin(i) / \sin(r)$ is ratio of speeds or top of ray slows down first and enters the block first | B1    |
| 3(a)(ii) | (wavelength) decreases                                                                                                                                                                    | B1    |
|          | (frequency) does not change.                                                                                                                                                              | B1    |
| 3(b)(i)  | $\sin(r) = \sin(i) / n$ or $\sin(45^\circ) / 1.6$ or $0.71 / 1.6$ or $0.44$ or $(r =) 21^\circ$                                                                                           | C1    |
|          | $26^\circ$                                                                                                                                                                                | C1    |
|          | $64^\circ$                                                                                                                                                                                | A1    |
| 3(b)(ii) | $64^\circ$ / angle of incidence / $i$ is greater than the critical angle / $39^\circ$ or angle of incidence = $64^\circ$                                                                  | B1    |
|          | it undergoes total internal reflection                                                                                                                                                    | B1    |

| Question | Answer | Marks |
|----------|--------|-------|
| 4(a)(i)  | 13 A   | B1    |
| 4(a)(ii) | 0 (A)  | B1    |

6 A dentist uses a plane mirror to see the back of a tooth.

(a) A plane mirror produces an image of an object.

Describe the position of this image.

.....  
.....  
..... [2]

(b) Fig. 6.1 shows the plane mirror used by the dentist to see the point labelled X on the tooth.

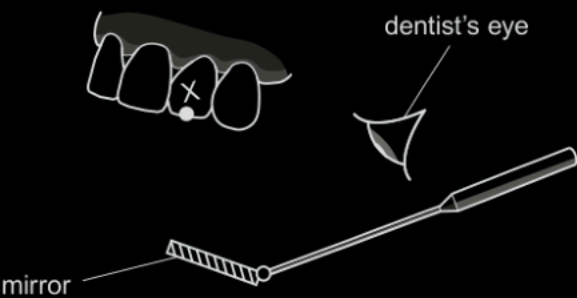


Fig. 6.1

On Fig. 6.1:

- (i) mark the position of the image of X formed by the mirror [1]  
(ii) draw a ray of light from X to show how the dentist can see the tooth. [2]

(c) State **one** characteristic of the image formed by the plane mirror other than its position.

.....  
..... [1]

[Total: 6]



## 7) 5054/21/M/J/20 Q6 — plane mirror image and ray construction — matching MS screenshot

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| Question | Answer                                                                                                          | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------|-------|
| 5(a)     | pressure = force / area                                                                                         | B1    |
|          | sharp knife has small(er) area and large(r) pressure<br>or blunt knife has large(r) area and small(er) pressure | B1    |
| 5(b)(i)  | temperature                                                                                                     | B1    |
|          | mass of gas or number of molecules                                                                              | B1    |
| 5(b)(ii) | molecules hit sides of container                                                                                | B1    |
|          | twice as many molecules per second hit (unit area)<br>or molecules hit twice as frequently                      | B1    |

| Question | Answer                                                          | Marks |
|----------|-----------------------------------------------------------------|-------|
| 6(a)     | behind the mirror or on the opposite side (of mirror) to object | B1    |
|          | same distance as object (from mirror)                           | B1    |
| 6(b)(i)  | image marked at correct position by eye                         | B1    |
| 6(b)(ii) | ray from tooth to mirror and ray from mirror to eye             | M1    |
|          | angles of incidence and reflection equal (by eye)               | A1    |
| 6(c)     | virtual or same size (as object) or laterally inverted          | B1    |



9 Some waves are longitudinal and some waves are transverse.

(a) State how a longitudinal wave differs from a transverse wave.

.....  
..... [1]

(b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

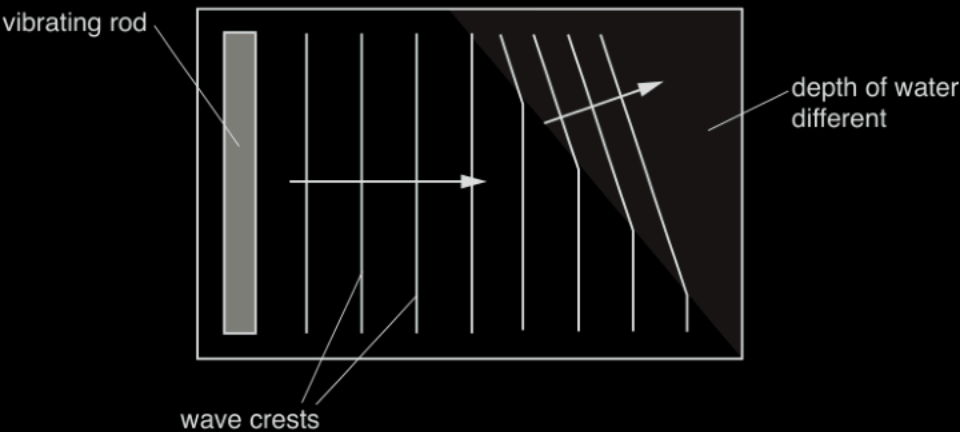


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

(i) In the left-hand section of the tank, the wavelength of the wave is 0.019 m and it travels at 0.17 m/s.

Calculate the frequency of the wave.

frequency = ..... [2]

(ii) State what happens to the frequency of the wave, as it passes into the right-hand section of the tank.

..... [1]

(iii) Using Fig. 9.1, state and explain what happens to the speed of the wave as it passes into the right-hand section of the tank.

.....  
.....  
..... [2]



(c) Fig. 9.2 shows light, in air, striking the vertical side of a rectangular glass block at an angle of incidence of  $60^\circ$ .

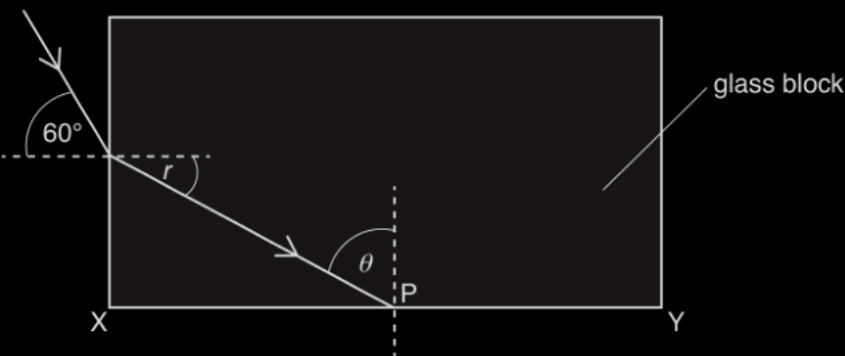


Fig. 9.2

The refractive index of the glass is 1.6. The light travels in the glass and strikes side XY at P.

(i) Underline all the terms that describe a light wave.

electromagnetic                      longitudinal                      transverse

[1]

(ii) At the point where the light enters the glass, the angle of refraction is  $r$ .

Calculate angle  $r$ .

$r =$  ..... [2]

(iii) 1. Calculate the critical angle  $c$  for light travelling in the block.

$c =$  ..... [2]

2. At P, the angle  $\theta$  between the ray and the normal is given by  $\theta = 90^\circ - r$ .

State and explain what happens to the light when it strikes side XY.

.....  
.....  
..... [2]

3. On Fig. 9.2, draw the path of the light after it strikes side XY at P and the path of the light when it is again travelling in the air. [2]



## 8) 5054/21/O/N/18 Q9 — ripple-tank refraction and light refraction — matching MS screenshot

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| Question | Answer                                                                                                                                         | Marks     |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 8(d)(ii) | from chemical (energy) <b>or</b> chemical (energy) as first term                                                                               | <b>B1</b> |
|          | to gravitational potential (energy) <b>or</b> to thermal / heat (energy) as last term                                                          | <b>B1</b> |
|          | to thermal / internal (energy) / heat <b>and</b> thermal / internal (energy) / heat                                                            | <b>B1</b> |
| 8(e)     | no resultant force <b>or</b> forces balance / cancel <b>or</b> tension equals weight (of bucket) <b>or</b> upwards force equals downward force | <b>B1</b> |

| Question  | Answer                                                                                                                                                                                                                                                  | Marks     |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 9(a)      | the <u>vibration</u> direction is parallel to the wave / energy travel direction<br><b>or</b> longitudinal waves have compressions and rarefactions <b>or</b> transverse waves have crests and troughs <b>or</b> longitudinal waves cannot be polarised | <b>B1</b> |
| 9(b)(i)   | $(f =) v / \lambda$ <b>or</b> 0.17 / 0.019                                                                                                                                                                                                              | <b>C1</b> |
|           | 8.9 Hz                                                                                                                                                                                                                                                  | <b>A1</b> |
| 9(b)(ii)  | (frequency) stays constant <b>or</b> does not change                                                                                                                                                                                                    | <b>B1</b> |
| 9(b)(iii) | (speed) decreases                                                                                                                                                                                                                                       | <b>B1</b> |
|           | (Fig. 9.1 shows that) wavelength decreases <b>or</b> refraction is towards normal<br><b>or</b> $i > r$ <b>or</b> top of wave (in shaded region) lags behind bottom of wave                                                                              | <b>B1</b> |
| 9(c)(i)   | electromagnetic <b>and</b> transverse underlined                                                                                                                                                                                                        | <b>B1</b> |
| 9(c)(ii)  | $n = \sin(i) / \sin(r)$ <b>or</b> $(r =) \sin^{-1}(\sin(i) / n)$ <b>or</b> $(r =) \sin^{-1}(\sin(60^\circ) / 1.6)$                                                                                                                                      | <b>C1</b> |
|           | 33(°)                                                                                                                                                                                                                                                   | <b>A1</b> |

9) 5054/21/M/J/17 Q10 — plane mirror, wavefront, reflection/refraction of waves, sound reflection — QP screenshot

- 10 (a) Fig. 10.1 shows two rays from an object that is placed in front of a plane mirror.

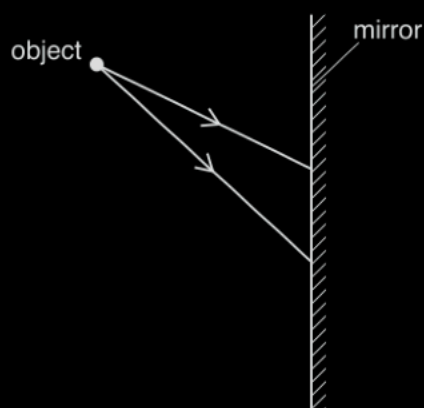


Fig. 10.1

- (i) On Fig. 10.1, draw the two reflected rays and locate the position of the image. [2]
- (ii) Apart from its position, state one characteristic of the image.

.....[1]

- (b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.



Fig. 10.2

- (i) State what is meant by *wavefront*.

.....

.....[2]

9) 5054/21/M/J/17 Q10 — plane mirror, wavefront, reflection/refraction of waves, sound reflection — QP screenshot



- (ii) Both reflection and refraction occur at the boundary.
1. On Fig. 10.2, draw the reflected wavefronts. [1]
  2. On Fig. 10.2, draw the refracted wavefronts. [2]
- (c) A loudspeaker produces a sound of frequency 2.0 kHz. The wavelength of this sound in air is 16 cm.
- (i) Calculate the speed of sound in air.

speed = .....[2]

- (ii) 1. State the range of frequencies that can be heard by a healthy human ear.  
.....[1]
2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength = .....[1]

- (iii) Describe a simple experiment to show that sound waves obey the law of reflection.  
You may draw a diagram if you wish.

.....

.....

.....

.....[3]

## 9) 5054/21/M/J/17 Q10 — plane mirror, wavefront, reflection/refraction of waves, sound reflection — matching MS screenshot

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| Question   | Answer                                                                                                   | Marks |
|------------|----------------------------------------------------------------------------------------------------------|-------|
| 10(a)(i)   | both reflected rays correct by eye                                                                       | B1    |
|            | image in correct position shown by continuation of rays behind mirror                                    | B1    |
| 10(a)(ii)  | virtual <b>or</b> upright <b>or</b> laterally inverted <b>or</b> same size (as object)                   | B1    |
| 10(b)(i)   | line ( joining points )                                                                                  | C1    |
|            | line joining points / particles on wave with same phase <b>or</b> line joining points along a crest etc. | A1    |
| 10(b)(ii)1 | correct angle to surface $\pm 7^\circ$ with correct orientation <b>and</b> similar wavelength (by eye)   | B1    |
| 10(b)(ii)2 | at least two lines at smaller angle to surface with correct orientation in slower medium                 | B1    |
|            | showing smaller and constant wavelength with wavefronts deviated in correct direction                    | B1    |
| 10(c)(i)   | $(v = ) f\lambda$ <b>or</b> $2(000) \times 16$                                                           | C1    |
|            | 32 000 cm / s <b>or</b> 320 m / s                                                                        | A1    |
| 10(c)(ii)1 | a range with 15–25 Hz as the lowest frequency <b>and</b> 15–30 kHz as the highest                        | B1    |
| 10(c)(ii)2 | 1.6 cm (if highest frequency is 20 000 Hz)                                                               | B1    |
| 10(c)(iii) | tube or other method to produce (narrow) beam of sound on source and/or on detector                      | B1    |
|            | stated detector <b>or</b> stated reflector                                                               | B1    |
|            | detector moved to find maximum loudness <b>or</b> angles $i$ and $r$ measured with suitable experiment   | B1    |

**9) 5054/21/M/J/17 Q10 — plane mirror, wavefront, reflection/refraction of waves, sound reflection — matching MS screenshot**

5054/21

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| Question  | Answer                                                                                                                                                                                                                                                                | Marks |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 11(a)     | battery and ammeter symbols correct                                                                                                                                                                                                                                   | B1    |
|           | thermistor symbol correct                                                                                                                                                                                                                                             | B1    |
|           | all connected in series                                                                                                                                                                                                                                               | B1    |
| 11(b)(i)1 | 0.05 (A) seen                                                                                                                                                                                                                                                         | C1    |
|           | $(V = ) IR$ or $0.05 \times 240$                                                                                                                                                                                                                                      | C1    |
|           | 12 V                                                                                                                                                                                                                                                                  | A1    |
| 11(b)(i)2 | $(R = ) V / I$ or 0.12 (A) seen                                                                                                                                                                                                                                       | C1    |
|           | 100 $\Omega$                                                                                                                                                                                                                                                          | A1    |
| 11(b)(ii) | description of change in resistance in two equal changes in temperature, e.g. 240–100 and 100–48                                                                                                                                                                      | C1    |
|           | difference values obtained are not the same, with one value shown e.g. 140 and 52 ( $\Omega$ )                                                                                                                                                                        | A1    |
| 11(c)(i)  | two different metals connected together                                                                                                                                                                                                                               | B1    |
|           | voltmeter, ammeter or galvanometer in series                                                                                                                                                                                                                          | B1    |
|           | clear junction of two different metals used as the place to measure temperature                                                                                                                                                                                       | B1    |
| 11(c)(ii) | ANY 2 lines from<br>measures high(er) temperatures<br>more sensitive to changes in temperature<br>measures fast(er) changing temperatures / reading can be taken more rapidly<br>produce electronic output / output may be stored or processed / may be read remotely | B2    |